

Invariance Constraints and the Non-Predictivity of Zero-Point Vacuum Energy in Quantum Field Theory

An Analysis of the Cosmological Constant Problem

Amos Jay Maley

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Abstract

The discrepancy between zero-point vacuum energy estimates in quantum field theory and the observed cosmological constant is often framed as a severe prediction failure. This paper argues that such estimates do not constitute standalone predictions of the theory. The argument is formulated in terms of invariance constraints: quantities that depend on regulator-dependent global offsets are not invariant under the admissible transformations that preserve the empirical witness ledger, and therefore do not define independent predictive observables.

A physics-bound AASC formulation is used to make these constraints explicit and to classify the roles available to vacuum-energy-like quantities. The formal primitives are introduced only through their standard physical correspondents: standing equivalence corresponds to the quotient by admissible scheme, coordinate, gauge, and normalization transformations; witnesses correspond to invariant measurable responses or prediction records; skin corresponds to convention-dependent coordinates; and boundary-only input corresponds to empirically fixed parameters not derived by the same interior theory. Within this formulation, absolute zero-point vacuum energy fails the required invariance criteria, while relational observables such as Casimir forces, phase differences, and response quantities remain well-defined.

The semiclassical gravitational coupling further demonstrates non-identifiability between finite vacuum-energy shifts and the cosmological counterterm. The result is a structured formulation and derivation of the non-predictivity of zero-point vacuum energy, consistent with renormalization and effective field theory. It removes the zero-point comparison as a valid prediction-forming procedure while leaving intact the unresolved naturalness, radiative-stability, and matching problem for the measured effective cosmological constant. The framework is used here only to make explicit invariance constraints already implicit in quantum field theory.

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1 Purpose and thesis

This paper is a physics-bound version of a full AASC closure result. Its purpose is not to ask the reader to accept a new formal language before seeing contact with established physics. The contact is stated first. In renormalized QFT, a homogeneous vacuum term proportional to the metric is not fixed as an independent prediction of the matter theory. It can be shifted by a finite local counterterm, and in semiclassical gravity the compensating shift of the cosmological term leaves the observed curvature coefficient unchanged. This is the familiar technical setting behind the claim that the cutoff zero-point sum is not a standalone prediction of QFT [1–5, 8, 9].

The contribution of the AASC formulation is to provide a structured formulation and derivation of this boundary as a general closure law. The derivation does not merely restate that one regularization scheme gives a different number than another. It classifies the attempted prediction-forming procedure. A quantity that changes under a same-scope redescription preserving all admitted witnesses is skin. A quantity that survives only as a difference, transition, response, or boundary-conditioned comparison is relational standing content. A quantity fixed only by matching to a larger envelope is boundary-only input. These roles exhaust the target. There is no fourth role in which the absolute zero of a single fixed vacuum interior becomes a standing-bearing scalar.

The result is narrow. It does not deny cosmic acceleration, the existence of a renormalized cosmological term in the gravitational effective action, curvature-dependent quantum effects, Casimir forces, or energy differences between dynamically connected phases. It also does not solve the technical-naturalness problem associated with the small measured value of Λ_{eff} . It removes only the claim that summing zero-point modes produces a failed cosmological prediction.

2 Early QFT anchor

The physics target is the standard zero-point expression

$$\rho_{\text{zp}}(\Lambda_{\text{UV}}) = \frac{1}{2} \int^{\Lambda_{\text{UV}}} \frac{d^3k}{(2\pi)^3} \sqrt{k^2 + m^2} \sim \frac{\Lambda_{\text{UV}}^4}{16\pi^2} + \dots \quad (1)$$

The controversial inference is not the calculation of a regulated quantity. It is the subsequent step in which the absolute value of Eq. (1) is treated as a predicted cosmological source.

Two standard freedoms isolate the problem. First, a Hamiltonian may be shifted by a constant,

$$H \longmapsto H + CI, \quad (2)$$

without changing transition frequencies, S-matrix elements, forces, or Heisenberg equations after the usual phase transport. Second, the renormalized stress tensor in curved-spacetime QFT is not unique;

locality, covariance, and conservation permit finite local conserved counterterm ambiguities [2–6]. Its homogeneous metric-proportional component may be written

$$\langle T_{\mu\nu} \rangle_{0,R} = -\rho_R g_{\mu\nu}, \quad (3)$$

with finite redefinition

$$\rho_R \longmapsto \rho_R + \delta\rho. \quad (4)$$

This is the restricted sector considered below. The theorem does not classify the full stress tensor in arbitrary curved backgrounds, where curvature-dependent counterterms, anomalies, state dependence, boundary terms, and material response matter separately.

In semiclassical gravity,

$$G_{\mu\nu} + \Lambda_R g_{\mu\nu} = 8\pi G_R \langle T_{\mu\nu} \rangle_R, \quad (5)$$

so a homogeneous matter-sector shift is compensated by

$$\rho_R \mapsto \rho_R + \delta\rho, \quad \Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho. \quad (6)$$

The invariant curvature coefficient is the total

$$\Lambda_{\text{eff}} = \Lambda_R + 8\pi G_R \rho_R, \quad (7)$$

with sign convention fixed by Eq. (5). Curvature observations do not independently identify the split into Λ_R and ρ_R . This is the immediate physics anchor for the AASC theorem.

3 Primitive correspondence with standard physical structures

The AASC terminology is used below only with the following operational reading. The table is part of the argument: it fixes how each primitive binds to standard physical practice.

AASC primitive	Physics correspondent	Role in this paper
Scope S	Fixed theoretical domain: here the homogeneous metric-proportional vacuum sector of QFT plus its semiclassical gravitational matching slot	Prevents overreach into the full stress tensor, anomalies, material Casimir modeling, or all cosmology
Presentation space P_S	A choice of regularization, subtraction convention, coordinates, fields, gauge, and parameter split	Contains calculational representatives, not all of which are observables
Standing equivalence $\sim_S$	Equivalence under admissible scheme, gauge, coordinate, and normalization transformations preserving fixed empirical records	The quotient that removes convention-dependent descriptions
Admissible redescription $\mathcal{R}_S$	Same-scope finite transformations preserving locality, covariance, conservation, and already fixed normalization conditions	Includes the homogeneous shift $\rho_R \mapsto \rho_R + \delta\rho$ with compensating Λ_R shift when gravity is included
Witness family $\mathcal{W}_S$	Invariant empirical outputs: transition frequencies, scattering data, forces, pressures, response functions, phase differences, curvature observations after parameter fixing	Determines what has physical standing inside the scope
Standing content	Equivalence class plus invariant witness ledger	The physical content left after quotienting away representation choices
Skin	Convention-dependent coordinate: gauge choice, zero of potential, subtraction constant, additive Hamiltonian zero, non-identifiable split of Λ_{eff}	Useful in calculation but not itself a standalone prediction
Boundary-only input	Empirically fixed parameter or cross-envelope matching datum not derived by the same interior theory	The measured Λ_{eff} is retained here; its smallness remains a naturalness and matching problem

This mapping is deliberately conservative. It does not claim that AASC replaces QFT. It states that, for the target sector, the AASC quotient and witness conditions instantiate the same invariance discipline already used in renormalized physics.

Proposition 1 (Binding consistency). *For the restricted homogeneous metric-proportional vacuum sector considered in this paper, AASC admissibility coincides with invariance under the finite renormalization and covariant stress-tensor redefinitions that preserve the already fixed empirical witness ledger. In particular, the transformation $\rho_R \mapsto \rho_R + \delta\rho$, together with the compensating semiclassical shift of Λ_R , is an admissible redescription exactly when it preserves the same transition data, response quantities, phase comparisons, and curvature records.*

Proof. Within the restricted sector, the only datum varied is the homogeneous metric-proportional coefficient of the vacuum stress tensor. The shift $\rho_R \mapsto \rho_R + \delta\rho$ changes no transition frequency, scattering record, local response function, Casimir force, phase-difference witness, or curvature

prediction once the compensating transformation $\Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho$ is made. Conversely, a transformation that changed any of these records would not preserve the witness ledger and therefore would not be an admissible same-scope redescription. Thus, in this sector, AASC admissibility and the standard finite-renormalization invariance condition select the same quotient. $\square$

In particular, the transformation $T_{\mu\nu} \mapsto T_{\mu\nu} + Cg_{\mu\nu}$, equivalently $\rho_R \mapsto \rho_R + \delta\rho$ in the homogeneous sector, preserves all invariant observables while shifting the vacuum-energy coordinate. This explicitly demonstrates that invariance under the admissible transformation class excludes absolute offsets from the standalone observable class.

4 Physics-native theorem

The following theorem is stated without AASC terminology. It is the physical result whose closure derivation is given afterward.

Definition 1 (Restricted homogeneous vacuum sector). Let $\mathcal{V}_0$ be the sector of scalar assignments depending only on the coefficient ρ_R of the homogeneous metric-proportional vacuum stress tensor in Eq. (3), on a fixed background class, with curvature-dependent counterterms, boundary/material counterterms, and non-vacuum matter normalization conditions fixed independently. The finite transformation group acting on this sector is

$$\mathcal{G}_0 : \quad \rho_R \mapsto \rho_R + \delta\rho, \quad (8)$$

where $\delta\rho$ is arbitrary and finite.

Theorem 1 (Physics form: no standalone absolute zero-point prediction). *No nonconstant scalar depending only on ρ_R in $\mathcal{V}_0$ is invariant under $\mathcal{G}_0$. Consequently, the absolute homogeneous matter-sector vacuum energy is not a standalone prediction of QFT.*

Proof. Let $X(\rho_R)$ be invariant under $\mathcal{G}_0$. Then

$$X(\rho) = X(\rho + \delta\rho) \quad \text{for every } \delta\rho \in \mathbb{R}. \quad (9)$$

The translation orbit of any ρ is all of $\mathbb{R}$. Hence X is constant on the quotient by $\mathcal{G}_0$. A constant on this quotient carries no information about the absolute value of ρ_R . Therefore the absolute matter-sector vacuum offset has no nonconstant invariant predictive content in the restricted homogeneous sector. $\square$

The theorem is algebraically simple because the obstruction itself is simple: the zero-point construction asks a convention-dependent coordinate to do the job of an invariant prediction. The content of the result is scope control. It identifies the maximal target sector in which the zero-point construction could produce an invariant cosmological prediction and shows that this sector excludes absolute vacuum energy.

Corollary 1 (Prediction-form obstruction). *Any procedure that assigns physical meaning to a regulator-dependent global offset without an invariant relational structure does not define a valid standalone prediction within the target theory.*

Proposition 2 (Non-identifiability in semiclassical gravity). *Within semiclassical gravity, no curvature observation distinguishes a finite shift of ρ_R from the compensating shift of Λ_R in Eq. (6). The identifiable coefficient is Λ_{eff} , not the split (Λ_R, ρ_R) .*

Proof. Substituting Eq. (3) into Eq. (5) shows that the homogeneous contribution enters only through Λ_{eff} as defined in Eq. (7). Transformation (6) leaves Λ_{eff} invariant. Since the curvature equation is unchanged, curvature observations cannot separate the two shifted pieces without an additional convention or new empirical input fixing the split itself. $\square$

5 Minimal AASC apparatus, physically bound

Only the local machinery needed to derive Theorem 1 is introduced. Each term should be read through the correspondence table in Section 3.

Definition 2 (Bookkeeping structure). For a fixed scope S , a bookkeeping structure is a tuple

$$\mathfrak{B}_S = (P_S, \sim_S, \mathcal{W}_S, \mathcal{R}_S, \mathcal{E}_S), \quad (10)$$

where P_S is the presentation space, $\sim_S$ is standing equivalence, $\mathcal{W}_S$ is the admitted witness-output family, $\mathcal{R}_S$ is the family of admissible redescrptions, and $\mathcal{E}_S$ is the envelope-boundary ledger.

Definition 3 (Standing content). The standing content of a presentation $p \in P_S$ is

$$\Sigma_S(p) = ([p]_S, \mathcal{W}_S([p]_S)), \quad (11)$$

where $[p]_S$ is its equivalence class under $\sim_S$. A datum has standing only when it is fixed by this quotient-and-witness pair.

Definition 4 (Admissible redescription). A map $r \in \mathcal{R}_S$ is an admissible redescription when it preserves the scope, the standing class, the admitted witness ledger, and the boundary status. Equivalently, it is a same-domain change of presentation rather than a change of physical content.

Definition 5 (Skin). A coordinate, label, normalization, gauge choice, subtraction convention, or additive zero is skin when it may be changed by an admissible redescription without changing Σ_S , the witness ledger, or the boundary ledger.

Definition 6 (Boundary-only input). A datum is boundary-only when it is required to instantiate or match the scope to an external envelope but is not fixed by same-scope admissibility. A boundary-only datum may be physical and measurable, but it is not an internal prediction of the occupant in that scope.

Definition 7 (Standing-bearing scalar). A scalar assignment X is standing-bearing in scope S only if for every admissible redescription $r \in \mathcal{R}_S$,

$$X(rp) = X(p) \quad (12)$$

after canonical transport of units and presentation conventions. If it changes under a witness-preserving same-scope redescription, it is skin, not standing content.

6 AASC closure theorem

Definition 8 (Ledger-offset redescription). Let L be a scalar ledger coordinate in scope S . A translation family

$$\tau_c : L \mapsto L + c \quad (13)$$

acts as a ledger-offset redescription when it preserves all admitted differences, derivatives, transition quantities, response functions, witness outputs, and boundary assignments.

Definition 9 (Absolute-offset candidate). An absolute-offset candidate is a scalar assignment A satisfying three conditions:

- (i) *single-interior form*: A is assigned to one fixed interior without a comparison, boundary, phase-transition, or response slot;
- (ii) *offset sensitivity*: there exists an admissible ledger-offset family τ_c such that $A(\tau_c p)$ depends nontrivially on c ;
- (iii) *standing claim*: A is asserted to be standing-bearing rather than skin or boundary input.

Lemma 1 (Zero-flux invariance). *If a redescription changes no standing class, no admitted witness output, and no boundary status, then every standing-bearing scalar is invariant under it.*

Proof. A standing-bearing scalar is fixed by the standing equivalence class and the witness ledger. If both are preserved, the transported scalar value is preserved under the equivalence relation. If it changed, the scalar would not descend to the standing quotient. $\square$

Lemma 2 (Common offsets are skin when witnesses are relational). *Suppose the admitted witness family depends on a scalar ledger coordinate L only through differences, derivatives, transitions, response quantities, or boundary-conditioned comparisons. Then every common translation $L \mapsto L+c$ is skin.*

Proof. Differences, derivatives, transitions, and response quantities are unchanged by a common additive translation. Boundary-conditioned comparisons are unchanged when the same convention is transported through the compared configurations. By hypothesis, no same-interior witness fixes the absolute zero of L . The common offset therefore changes only the presentation coordinate and not the standing content. $\square$

Theorem 2 (AASC form: null-offset closure). *No non-relational absolute-offset candidate can be standing-bearing in an admissible AASC bookkeeping structure unless an admitted same-interior witness fixes the offset itself.*

Proof. Let A be an absolute-offset candidate, and suppose no admitted same-interior witness fixes its absolute zero. Then the relevant ledger-offset redescription τ_c preserves all admitted witnesses by Lemma 2. It is skin and changes no standing content. By Lemma 1, every standing-bearing scalar must be invariant under τ_c . But A changes nontrivially under τ_c by definition. Therefore A cannot be standing-bearing. The only escape is to add a witness that fixes the offset itself, in which case the target is no longer witness-free absolute offset. $\square$

In the physics correspondence fixed above, Theorem 2 states that no renormalization-invariant observable can be constructed from a global additive offset to the Hamiltonian or the homogeneous metric-proportional component of the stress tensor.

Corollary 2 (AASC vacuum-energy boundary). *An absolute vacuum-energy scalar assigned to a single fixed vacuum interior has no standing in an admissible AASC structure in the absence of an admitted same-interior witness fixing the absolute zero.*

7 Equivalence between the physics theorem and the AASC closure theorem

The physics theorem and the AASC theorem are not two unrelated arguments. They are the same closure under the correspondence in Section 3.

Proposition 3 (Physics-to-AASC binding). *Under the correspondence table, the finite transformation $\rho_R \mapsto \rho_R + \delta\rho$ is a ledger-offset redescription, the cutoff zero-point assignment is an absolute-offset candidate when treated as a single-interior cosmological source, and the invariant QFT observables are the admitted witness family.*

Proof. The finite shift changes the scalar coordinate ρ_R while preserving the non-vacuum matter normalization conditions, local covariance and conservation constraints, and curvature predictions after the compensating Λ_R shift. It therefore has the form of a witness-preserving ledger-offset redescription. A zero-point assignment treated as a direct cosmological source is assigned to a single fixed vacuum interior, changes under this shift, and is asserted to be predictive. It is therefore an absolute-offset candidate. Transition frequencies, scattering data, response quantities, Casimir forces, phase differences, and curvature observations of Λ_{eff} are precisely the invariant witnesses in the target scope. $\square$

Theorem 3 (Projection theorem). *The QFT non-predictivity result is recovered as the physical realization of Theorem 2 under the correspondence defined in Proposition 3. Conversely, Theorem 2 supplies the general closure reason why Theorem 1 holds in the restricted homogeneous sector.*

Proof. By Proposition 3 and the binding consistency result in Proposition 1, the QFT homogeneous shift is an instance of ledger-offset redescription and the zero-point vacuum-energy assignment is an absolute-offset candidate. Theorem 2 says such a candidate cannot be standing-bearing without a same-interior witness fixing the offset. In QFT language, this says no nonconstant scalar depending only on ρ_R is invariant under $\rho_R \mapsto \rho_R + \delta\rho$, which is exactly Theorem 1. Conversely, the QFT theorem instantiates the general AASC rule in the restricted homogeneous metric-proportional sector. $\square$

8 Exhaustion, no repair, and surviving physics

Theorem 4 (Role exhaustion for vacuum-energy-like quantities). *Let V be any vacuum-energy-like scalar construction in a fixed scope satisfying the correspondence in Section 3. If V is admissible, then it falls into exactly one of the following roles:*

- (i) **skin**: an absolute offset changed by a witness-preserving redescription;
- (ii) **relational standing content**: a difference, derivative, response, transition, phase comparison, or boundary-conditioned quantity whose witnesses are invariant under common offsets;
- (iii) **boundary-only input**: an empirically fixed cross-envelope parameter not internally derived by the same occupant.

There is no fourth admissible role corresponding to an independently standing absolute zero-point energy of a single fixed vacuum sector.

In physical terms, these three roles correspond respectively to scheme-dependent normalizations, invariant response observables such as Casimir forces or phase comparisons, and empirically fixed parameters such as the renormalized effective cosmological constant. This classification is exhaustive with respect to quantities invariant under the admissible transformation class defined in Sections 3–5.

Proof. If V changes under a witness-preserving common offset, Lemma 1 excludes it from standing content; it is skin. If V is invariant because it is built from comparisons, derivatives, transitions, responses, or phase relations, its standing is carried by the relational witness slot. If V is not fixed by same-interior admissibility but is fixed by external cosmological matching, it is boundary-only input. The standing quotient, witness ledger, and boundary ledger exhaust the roles available inside the scope. Therefore no fourth role remains. $\square$

Proposition 4 (No same-scope repair). *No same-scope construction, ultraviolet completion, cancellation mechanism, or selection rule converts the zero-point summation into a standalone invariant prediction unless it adds an admitted witness, changes the scope, or makes the datum boundary-only. Such mechanisms may still address the value, stability, or matching of the renormalized cosmological parameter.*

Proof. A same-scope admissible construction that preserves the witness ledger and boundary ledger has zero standing flux and cannot turn skin into standing content by Lemma 1. If it adds a witness, the target has changed because the absolute offset is no longer witness-free. If it changes the domain in which the quantity is evaluated, the scope has changed. If it fixes the value by external empirical matching, the datum is boundary-only rather than internally predicted. These alternatives exhaust the admissible repairs. $\square$

8.1 Casimir forces and material response

Casimir forces survive because they are relational response quantities, not witnesses of an absolute bulk offset. Schematically,

$$F(a) = -\frac{\partial}{\partial a}(E(a) - E(a_0)), \quad (14)$$

so a common additive offset cancels before the measured force is obtained. This statement does not depend on idealized perfect-conductor calculations. In realistic material systems the measured force is a renormalized response of a specified apparatus and remains invariant under shifts of the global energy zero. This is consistent with the view that Casimir forces need not be interpreted as direct measurements of bulk zero-point substance [10–13].

8.2 Phase transitions, metastable vacua, and tunneling

The theorem does not remove energy differences between distinct phases or dynamically connected vacua. Phase differences, tunneling rates, domain-wall dynamics, and transition actions depend on relational structure and are therefore role (ii) in Theorem 4. What is excluded is the assignment of independent standing to the absolute homogeneous offset of one fixed vacuum without a comparator or boundary witness [14, 15].

8.3 Naturalness and matching remain

The measured Λ_{eff} remains an empirical gravitational parameter. Its smallness relative to particle-physics scales is still a naturalness, radiative-stability, and matching problem for EFT and gravity

[1, 7, 9]. The closure theorem removes only the stronger claim that a regulator-dependent zero-point sum, considered by itself, is a failed cosmological prediction.

9 Final projection into QFT

The abstract closure now returns to standard physics in one line of projection:

$$\begin{aligned} &\text{standing-preserving offset redescription} \\ &\Downarrow \\ &\rho_R \mapsto \rho_R + \delta\rho, \quad \Lambda_R \mapsto \Lambda_R - 8\pi G_R \delta\rho, \quad \Lambda_{\text{eff}} \text{ fixed.} \end{aligned} \tag{15}$$

The cutoff zero-point sum is a coordinate on the left side of this projection, not an invariant on the right. Casimir forces and phase differences survive because their witness structure is relational. Curvature observations determine Λ_{eff} , not the split into matter-sector vacuum energy and a cosmological counterterm. Therefore QFT does not admit the standalone prediction usually attributed to Eq. (1).

This is the precise point at which the common many-orders-of-magnitude comparison leaves the predictive content of the theory. The comparison is not a failed prediction; it is a failed prediction-forming procedure.

10 Conclusion

The physics-bound AASC version shows how the orthodox QFT boundary follows from the invariance conditions defined above. In physics language, the homogeneous matter-sector vacuum offset is shifted by finite renormalization freedom and is not independently identifiable from the cosmological counterterm. In the structural language used here, it is a non-relational absolute offset that changes under a witness-preserving redescription. Such a quantity is skin unless an admitted same-interior witness fixes it.

The result exhausts the target class. Vacuum-energy-like quantities are either convention-dependent offsets, relational witnesses, or boundary-only empirical inputs. No same-scope repair admits an invariant construction of an independently standing absolute zero-point energy. The observed effective cosmological constant remains physical and unresolved as a naturalness and matching problem. What is eliminated is only the inference that summing zero-point modes produces a standalone cosmological prediction of QFT.

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